

Activity 7

1. Before: `SELECT * FROM common_player_info WHERE display_first_last = 'LeBron James';`
After: `CREATE INDEX idx_player_name ON common_player_info(display_first_last);`

```
sqlite> SELECT * FROM common_player_info WHERE display_first_last = 'LeBron James';  
Run Time: real 0.043657 user 0.000000 sys 0.015625  
sqlite> CREATE INDEX idx_player_name ON common_player_info(display_first_last);  
Run Time: real 0.015554 user 0.000000 sys 0.000000  
sqlite> SELECT * FROM common_player_info WHERE display_first_last = 'LeBron James';  
Run Time: real 0.000381 user 0.000000 sys 0.000000
```

99.13% improved

2. Before: `SELECT t.full_name, COUNT(*) AS total_drafted FROM team t JOIN draft_history d ON t.id = d.team_id GROUP BY t.id ORDER BY total_drafted DESC;`

```
Run Time: real 0.064723 user 0.015625 sys 0.031250
```

After: `CREATE INDEX idx_draft_team_id ON draft_history(team_id);`

```
Run Time: real 0.023657 user 0.015625 sys 0.000000
```

63.4% Improvement

3. Before : `SELECT * FROM game WHERE team_id_home = '1610612744';`

```
Run Time: real 60.760636 user 2.328125 sys 14.734375  
sqlite> CREATE INDEX idx_game_team_id_home ON game(team_id_home);  
Run Time: real 0.093034 user 0.031250 sys 0.046875  
sqlite>
```

After: `CREATE INDEX idx_game_team_id_home ON game(team_id_home);`
`SELECT game_id, game_date, team_name_home, pts_home FROM game WHERE team_id_home = '1610612744';`

```
Run Time: real 11.153269 user 1.328125 sys 1.656250  
sqlite>
```

81.6% Improvement